Jobs Lost, marriages gained: The dual effects of immigration in Australia

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Abstract

This paper examines the effects of immigration on local labor market and marriage market outcomes in Australia. Using data from the 1991–2021 Australian Bureau of Statistics censuses and a shift-share instrumental variable approach within a fixed-effects framework, we estimate the causal impact of immigration at the local government area level. To assess the identifying variation underlying the instrument, we apply the Rotemberg weight decomposition. We find that immigration significantly increases the overall unemployment rate and high-skill employment, while reducing middle-skill and low-skill employment. Turning to marriage market outcomes, immigration exerts a positive and significant effect on the local marriage share, a finding that persists despite the concurrent labor market displacement. Gender-disaggregated results reveal a larger positive effect for females than for males. Taken together, these findings indicate that immigration reshapes local labor and marriage markets in ways that vary systematically across skill groups, gender, and the cultural composition of immigrant inflows.

Keywords: Immigration, labor market, marriage, enclave, heterogeneity analysis

JEL Classification: J61, J12, J21, R23

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1 Introduction

The relationship between immigration and labor market outcomes has long been a subject of debate in economics. A prominent strand of the literature emphasizes substitution between immigrants and native workers, suggesting that immigration may exert downward pressure on native employment or wages in local labor markets (Borjas, 2003; Dustmann et al., 2017; Monras, 2020). In contrast, a growing body of research highlights the importance of heterogeneity across skill groups, industries, and local labor market characteristics, documenting circumstances under which immigrants and native workers are instead complements in production (Card, 2001; Ottaviano and Peri, 2012; Foged and Peri, 2016). Empirical evidence on these issues has been drawn predominantly from the United States and European countries (Abramitzky and Boustan, 2017; Foged and Peri, 2016). However, despite Australia being one of the major immigrant-receiving countries among OECD membersand having experienced a 22.3 % increase in immigration relative to 2019 by 2024 – systematic empirical evidence on the labor market effects of immigration in the Australian context remains relatively scarce (OECD, 2025).

This paper examines the causal effects of immigration on labor market outcomes and marriage shares in Australia, using census data from the Australian Bureau of Statistics (ABS) spanning 1991 to 2021. We analyze how the share of immigrants within a local government area (LGA) affects unemployment rates, skill-specific employment rates–disaggregated by high-, middle-, and low-skill categories– and marriage rates among young adults, exploiting variation in immigration inflows across LGAs over time. To identify causal effects, we employ a fixed effects instrumental variable (FE-IV) approach with a shift-share IV following Card (2009) and Tabellini (2020), which allocates national-level immigrant inflows to LGAs based on the historical settlement patterns of immigrants from each origin country prior to Australia's immigration policy reform in 1996, when the system transitioned from a family-based to a skill-based selection framework (Larsen, 2013; Kim and Wang, 2024). Immigration inflows are not randomly distributed across regions but instead reflect migrants' endogenous location choices in response to local economic conditions, labor demand, and existing migrant networks; our instrument addresses this endogeneity by exploiting the predetermined 1991 immigrant distribution as a baseline that predates the policy reform and is therefore plausibly exogenous to contemporaneous local labor market shocks.

We go beyond the standard shift-share identification strategy by explicitly examining the sources of identifying variation underlying the instrument. Recent methodological advances have raised important concerns about the validity of shift-share instruments, particularly regarding whether historical settlement patterns may themselves reflect persistent local economic conditions, potentially violating the exclusion restriction (Jaeger et al., 2018; Goldsmith-Pinkham et al., 2020). To address this, we incorporate the Rotem-

berg decomposition proposed by Goldsmith-Pinkham et al. (2020), which reinterprets the shift-share IV as a generalized method of moments estimator that pools just-identified estimates, each using the settlement share of a single origin country as a separate instrument. By analyzing Rotemberg weights and the heterogeneity of the associated just-identified estimates across origin countries, we assess whether our findings are driven by a narrow set of source countries whose historical settlement patterns may be endogenous, thereby strengthening the credibility of our causal interpretation in a transparent and systematic manner.

Our empirical approach also addresses a key methodological limitation of the existing literature. Most studies that examine skill-specific labor market effects rely on the national skill-cell approach, which assigns immigrants to education-experience cells and exploits variation in immigrant inflows across these cells (Borjas et al., 1997; Borjas, 2003; Ottaviano and Peri, 2012). As demonstrated by Dustmann et al. (2016), however, this approach is susceptible to misclassification bias arising from immigrant downgradingthe well-documented tendency of immigrants to take jobs below their measured qualification level upon arrival. This bias is particularly acute for female immigrants, who face a compounded disadvantage in host country labor markets and are systematically more likely to experience occupational downgrading than their male counterparts (Akgüç and Parasnis, 2023; Akresh, 2006), implying that the misclassification bias is itself heterogeneous across gender. By adopting a pure spatial approach at the LGA level, our analysis does not require the pre-assignment of immigrants to skill groups and is therefore immune to this source of bias. Crucially, while we do not assign immigrants to skill groups, we exploit variation in the skill composition of employment outcomes as dependent variables, allowing us to recover skill-specific total effects of immigration without imposing ex ante assumptions about which workers compete directly with immigrants. The gender asymmetry in downgrading further motivates our disaggregated analysis of employment outcomes across both skill groups and gender.

Our analysis yields three sets of findings. First, we find that an increase in the immigration share raises the overall unemployment rate and generates heterogeneous effects across the skill distribution: high-skill employment increases while middle- and low-skill employment decline, consistent with a pattern of occupational upgrading among native workers alongside displacement at the lower end of the skill distribution. Second, these effects differ systematically by gender. The negative effect on low-skill employment and the positive effect on high-skill employment are both larger in magnitude for female workers, while the adverse unemployment effect is more pronounced for males. The negative effect on middle-skill employment is statistically insignificant for males but significant for females, suggesting that female middle-skill workers are more exposed to immigration-induced labor market pressures. Third, we find that immigration raises marriage rates among young adults aged 20-34, with the effect being substantially larger for females than

for males. Heterogeneity analysis further reveals that the positive marriage market effects of immigration are primarily concentrated in LGAs with a higher share of immigrants from non-English-speaking countries.

This paper makes three contributions to the existing literature. First, we contribute to the literature on the labor market impacts of immigration by providing regional-level evidence from Australia on the skill-specific employment effects of immigration. Unlike most existing studies that focus on wage outcomes or adopt an individual-level perspective, we examine the employment margin at the local level, which captures how immigration reshapes the overall structure of local labor markets rather than simply the relative outcomes of native versus immigrant workers. By adopting a pure spatial approach, our analysis recovers total employment effects that incorporate complementarities across skill groups and between labor and capital, while remaining immune to the misclassification bias inherent in the national skill-cell framework. Second, we contribute to the growing literature on the heterogeneous effects of immigration by providing systematic evidence on the joint skill-gender heterogeneity of immigration's employment effectsa dimension that has received limited attention in the regional-level literatureand connecting this heterogeneity to the gender asymmetry in immigrant downgrading. Third, we extend the analysis of immigration's socioeconomic consequences beyond the labor market by examining its effects on marriage rates, complementing the recent work of Carlana and Tabellini 2025 and Autor et al. 2019 in the historical U.S. context and providing the first regional-level evidence on immigration and marriage market outcomes in modern Australia. Our findings suggest that immigration reshapes not only the labor market but also the demographic structure of local communities, with distributional consequences that vary systematically across skill groups and gender.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 4.1 and Section 4.1 describes the data and variable construction. Section 4.2 and Section 4.3 presents the empirical framework, including the shift-share IV design and the Rotemberg decomposition. Section 5.1 reports the main results on labor market outcomes and marriage rates. Section 5.2 presents heterogeneity analyses by gender and English-speaking background. Section 6 reports robustness checks, and Section 7 concludes.

2 Related literature

The labor market consequences of immigration have been extensively studied, with findings that vary substantially across empirical speicifications, context, and skill groupds. Early influential work by Borjas (2003) exploits variation acorss national education-experience cells and finds that a 10% increase in labor supply reduces native wages by pproximately 4%, particularly for low-skilled workers. In contrast, studies exploiting

sptial variation in immigrant inflows tend to ind more modest effects. Card (2001) uses the Mariel boatlift as a natural experiment and finds no significant wage or employment effects on native worekrs in Miami, while Card (2005) documents small and heterogeneous impacts acorss U.S. cities. Monras (2020) exploits the 994-1995 Mexican peso crisis and finds significant displacement effects on low-skilled native workers, highlighting the importance of labor demand conditions on shaping the response to immigration shocks.

A separate strand of the literature emphasizes immigrant-native complementarities in production. Peri and Sparber (2009) show that immigrants disproportionately specialize in manual and physically intensive tasks, inducing native workers to reallocate toward communication-intensive occupations and generating wage gains at the upper end of the skill distribution. Ottaviano and Peri (2012) extend this framework to allow for imperfect substitutability between immigrants and natives within skill cells, finding that immigration raised average native wages over the period 19902006 in the United States. Using longitudinal employer-employee data from Denmark, Foged and Peri (2016) find that refugee inflows induced native workers to upgrade their occupational complexity, producing earnings gains particularly for low-skilled natives. Together, these studies suggest that the direction and magnitude of immigration's labor market effects depend critically on the degree of task complementarity and the local skill composition.

Dustmann et al. (2016) provide a unifying framework showing that the national skill-cell approach, the pure spatial approach, and the mixture approach estimate conceptually different parameters of the same canonical model. A key limitation of the national skill-cell approach is its susceptibility to misclassification bias arising from immigrant downgrading. Dustmann et al. (2016) and Akgüç and Parasnis (2023) document that immigrants systematically take jobs below their measured qualification level upon arrival, with the degree of downgrading varying substantially by gender and origin country. The pure spatial approach, by contrast, does not require the pre-assignment of immigrants to skill groups and therefore recovers total employment effects that incorporate complementarities across skill groups and between labor and capital (Card, 2001; Peri and Sparber, 2009).

The endogeneity of local immigration shares has been addressed in a large literature using shift-share instrumental variables following Card (2001). Tabellini (2020) employs this approach to study the political and economic consequences of European immigration to U.S. cities in the early twentieth century, finding positive effects on native employment and output. Jaeger et al. (2018) raise an important concern, showing that if settlement patterns are stable over time, the instrument may conflate short- and long-run responses to immigration shocks. Goldsmith-Pinkham et al. (2020) provide a complementary diagnostic framework, reinterpreting the shift-share IV as a GMM estimator and proposing Rotemberg weights to assess the sensitivity of the overall estimate to the exclusion re-

striction for each origin country. These methodological tools have been widely adopted in recent empirical work (Foged and Peri, 2016; Tabellini, 2020) but remain underutilized in the Australian immigration literature.

A growing literature examines gender-specific labor market responses to immigration. Cortes (2008) finds that low-skilled immigration significantly reduces the price of household services, inducing high-skilled native women to increase their labor supply and work longer hours. Foged and Peri (2016) document that occupational upgrading in response to refugee inflows is more pronounced among female native workers than males, suggesting that gender shapes both the competitive pressure and the complementarity gains from immigration. Despite these contributions, systematic evidence on the joint skill-gender heterogeneity of immigration's employment effectsparticularly at the local labor market levelremains limited.

Beyond labor markets, a growing body of research examines the broader socioeconomic consequences of immigration at the local level. Autor et al. (2019) show that trade-induced labor market shocks reshape household formation decisions, reducing marriage rates among affected workers. Angrist (2002) documents that immigration-induced shifts in local sex ratios significantly alter marriage market outcomes. Farnham et al. (2011) further show that changes in local housing prices affect marital stability, suggesting that immigration may also shape marriage decisions through its effects on local housing markets (Moallemi and Melser, 2020). The closest paper to ours is Carlana and Tabellini (2025), who use a shift-share design across U.S. cities in the early twentieth century and find that immigration raised native marriage and fertility rates. Despite these contributions, the socioeconomic consequences of immigration beyond the labor marketand marriage outcomes in particularremain largely unexplored in the Australian context.

3 Data

We use census-based community profiles at the local government area (LGA) level, collected at a quinquennial frequency by the Australian Bureau of Statistics (ABS). These profiles are constructed using the full census enumeration and include, among other variables, information on the number of foreign-born residents, the number of native-born residents, and a range of LGA-level labor market and demographic outcomes. Beginning in 2001, the ABS started releasing time-series community profiles that compile information across three consecutive census waves.

For example, the 2001 time-series community profiles provide LGA-level data for the census years 1991, 1996, and 2001. We use these profiles to construct variables that are consistently defined across LGAs over the period from 1991 to 2021. Specifically, the ABS Community profiles provide information on the number of immigrants residing

in each LGA by country of origin, which forms the basis of our instrumental variable. Our analysis examines the effects of immigration on local labor market and demographic outcomes, including unemployment rate, skill-specific employment rates, and marriage share at the LGA level. We construct explanatory variables using 1991 as the base-year population denominator, defined as the total LGA population in 1991. This choice is motivated by the structure of our shift-share instrumental variable, which uses 1991 as the base year for the initial settlement share. Following the shift-share literature (e.g., Card (2001); Peri (2012)), we allow for a lag between the base year and the start of the estimation sample to ensure that the initial settlement patterns predate the period of analysis and are thus less susceptible to concerns about reverse causality. Consequently, the effective estimation sample begins in 1996, which is the first period t for which these shares can be defined. The final estimation sample therefore consists of a balanced panel of 496 LGAs observed from 1996 to 2021.

To construct the key explanatory variable, the immigration share, we define immigrants as individuals born outside of Australia territory, identified through country-of-birth or birthplace information from the ABS data. Across the entire sample period, we consolidate all reported birthplaces into 29 origin groups to ensure comparability over time because the set of birthplaces recorded in the census varies across waves. These groups consist of 28 major countries and one residual category that aggregates all remaining, less consistently observed birthplaces. We then define the immigrant share in a county as the number of immigrants in the current year divided by the countys total population in 1991.

Our outcome variables comprise three sets of measures, all constructed at the LGA level from raw census counts. First, unemployment rates are defined as the number of unemployed individuals divided by the corresponding labor force. We construct three measures: the overall unemployment rate, and gender-specific rates for males and females, where the denominator is the male and female labor force, respectively. Second, skill-specific employment rates are defined as the number of employed individuals in each occupational category divided by the total labor force. We classify occupations into three skill tiers – high-skill, middle-skill, and low-skill – following the skill level classifications provided by the ABS. Third, the share of married individuals is constructed for the population aged 20-34, reflecting marriage behavior at prime marriage ages. For each of the three measures overall, male, and female the numerator is the number of married individuals in the respective group aged 20-34, and the denominator is the total population of the corresponding group in the same age range.

Table 1 reports the descriptive statistics for the main estimation sample, comprising 496 LGAs observed over six census year from 1996 to 2021, for a total 2,976 observations. The immigration share averages 26.4% across LGAs, with a standard deviation of 21.5%, reflecting considerable heterogeneity in immigrant concentration across local areas. The

mean unemployment rate is 6.1% overall, with male and female rates of 6.4% and 5.7%, respectively, indicating a modest but consistent gender gap across LGAs. Turning to skill-specific employment rates, the average high-skill employment rate is 35.1%, which exceeds both the low-skill rate (30.2%) and the middle-skill rate (26.6%). Gender-disaggregated employment rates reveal notable differences across skill tiers: female workers are more concentrated in middle-skill occupations (28.8%) relative to males (24.5%), while males exhibit higher employment rates in low-skill occupations (31.6%) compared to females (28.7%), with high-skill employment rates broadly similar across genders. Turning to marriage outcomes, women exhibit a higher average marriage share (36.4%) than men (26.7%). The number of observations for marriage outcomes is slightly smaller than for other variables, as LGAs with no residents in the 20-34 age group are assigned missing values for these measures.

[Table 1]

4 Methodology

4.1 Model

To examine the effects of immigration on local labor market outcomes – including unemployment rates and skill-specific employment rates – and demographic outcomes, namely marriage shares, we consider the following model:

$$y_{it} = \beta Immishare_{it} + X'_{i,t-1}\gamma + \rho_i + \phi_t + \varepsilon_{it} \tag{4.1}$$

where dependent variable, y_{it}, denotes the unemployment rate, skill-specific employment rates, or marriage share in LGA i in year t. The main explanatory variable, $ImmiShare_{it}$ is the share of immigrants in LGA i in year t. We control for lagged LGA-level socio-demographic characteristics $\mathrm{X}_{i,t-1}$ including the share of college graduates, to account for differences in human capital composition and demographic structure across local areas that may independently affect labor market and marriage outcomes. LGA fixed effects ρ_i absorb time-invariant local characteristics, such as geographic and institutional factors, while census-year fixed effects ϕ_t control for aggregate macroeconomic shocks common to all LGAs in a given period, and ε_{it} is an idiosyncratic error term. The coefficient of interest, β, measures the impact on each outcome variable of a one percentage-point rise in the immigration share. Whether β in equation (4.1) admits a causal interpretation depends critically on the exogeneity of local immigration share variation with respect to labor market and demographic outcomes, given the controls included in the specification. We proceed as follows: we first outline how the immigration share

measure is constructed, before turning to the identification strategy we employ to isolate the causal effect of immigration exposure on local labor market and marriage outcomes.

4.2 Identification strategy

We construct the immigration share, $ImmiShare_{it}$, as the total number of immigrants residing in LGA i in year t, divided by the total population of the LGA in 1991. Letting M_{ikt} denote the number of immigrants from origin country k residing in LGA i in year t. $ImmiShare_{it}$ can be written as:

$$ImmiShare_{it} = \frac{1}{pop_{i,1991}} \sum_{k=1}^{K} M_{ikt} = \frac{1}{pop_{i,1991}} \sum_{k=1}^{K} \frac{M_{ikt}}{M_{kt}} M_{kt}, \tag{4.2}$$

As shown in equation (4.2), the immigration share decompose naturally into two components: the national-level stock of immigrants from origin country k in year t (i.e., $\mathrm{M}_{k,t}$, the shift component), and the share of those national immigrants living in LGA i (i.e., $\mathrm{M}_{i,k,t}/\mathrm{M}_{k,t}$, the share component). Identifying the causal effect of $ImmiShare_{it}$ on local outcomes requires that its variation be exogenous to unobserved factors driving changes in unemployment and marriage outcomes at the LGA level. However, this assumption is unlikely to hold in practice, as the two components of equation (4.2) raise distinct endogeneity concerns.

First, the shift component, M_{kt}, is primarily determined by aggregate push factors in origin countries and by national immigration policies, rather than by local economic conditions in any single LGA. Consequently, changes in unemployment or marriage outcomes in a given LGA are unlikely to influence origin-specific national immigration stocks. For this reason, M_{kt} can be regarded as exogenous to local labor market and demographic conditions at the LGA level.

However, the second component, $\mathrm{M}_{i,k,t}/\mathrm{M}_{k,t}$ is potentially problematic. Changes in local labor market condition or marriage patterns within in an LGA may influence the composition of immigrant group who choose settle there, giving rise to endogeneity in this component. While LGA fixed effects in equation (4.1) partially mitigate this concern by absorbing time-invariant local characteristics, the presence of transitory local shocks remains a source of bias. Specifically, short-run economic shocks, changes in local amenities, or fluctuations in employment opportunities may simultaneously affect local outcomes and the sorting of immigrants across LGAs.

To address this endogeneity concern, we replace the contemporaneous share component with its pre-determined counterpart from 1991, prior to the start of our sample

period, yielding a shift-share instrument of the following form:

$$ImmiShare_{it}^{IV} \equiv \frac{1}{pop_{i,1991}} \sum_{k=1}^{K} \frac{M_{ik,1991}}{M_{k,1991}} M_{kt}, \tag{4.3}$$

This instrument takes the form of a weighted sum across origin countries, where the weights are given by the pre-determined settlement shares, $\frac{M_{ik,1991}}{M_{k,1991}}$, defined as the share of origin country k immigrants residing in LGA i in the base year 1991. These settlement shares are then applied to the national-level stock of immigrants from country k in year t, M_{kt}, to generate a predicted immigration share for each LGA-year cell. Intuitively, $ImmiShare_{it}^{IV}$ captures the immigration inflow that LGA i would be expected to receive in year t based solely on its historical settlement patterns, thereby isolating variation in the local immigration share that is driven by national-level changes in immigration rather than by contemporaneous local conditions.

We select 1991 as the base year for the settlement shares for several reasons following Kim and Wang (2024). First, it is the earliest census year for which detailed country of origin information is publicly available in the ABS data. Second, it predates the major immigration policy reforms of the mid-1990s, including the introduction of the Temporary Work (Skilled) visa in 1996, which marked a structural shift toward a more skill-oriented immigration system. Third, immigrant settlement patterns in 1991 predate the pronounced changes in the country of origin composition of immigrants documented since the 1990s (Miller, 1999), and using a pre-sample base year alleviates concerns that local immigrant composition is endogenous to subsequent economic conditions.

To establish the relevance of this instrument, consider each component of Z_{it} in equation (4.3) and compare it with the right-hand side of equation (4.2). First, both share the same origin-specific national immigrant stocks in year t, M_{kt}, which can be regarded as exogenous for the reasons outlined above. Second, rather than relying on contemporaneous settlement shares, $\frac{M_{ik,1991}}{M_{k,1991}}$, the instrument exploits the pre-determined settlement share of each immigrant group, $\frac{M_{ik,1991}}{M_{k,1991}}$, within an LGA. The predictive power of historical settlement patterns for subsequent location choices stems from a well-established tendency for immigrants to concentrate in areas that already host sizable communities from the same country of origin, a pattern first documented by Bartel (1989).

The validity of the instrument requires that ε_{it} in equation (4.1) be uncorrelated with the factors that shaped immigrants' settlement choices in 1991. We argue that this condition is satisfied with two reasons. First, local economic or demographic conditions within a single LGA are unlikely to drive national-level immigration trends, which rules out any direct feedback from local outcomes to the shift component, M_{kt}. Second, the substantial time elapsed between the base year settlement patterns and subsequent changes in local outcomes, together with the conditioning on LGa and year fixed effects in equation (4.1),

renders it implausible that ε_{it} is systematically related to the geographic distribution of immigrants in 1991. We further examine potential threats to the validity of this instrument in Section 6.

4.3 Rotemberg weights

Our instrumental variable follows the Bartik shift-share design of Card (2001), which interacts the historical distribution of immigrant across local government areas by origin country—the share component—with the subsequent growth in the national stock of immigrants from each origin country—the shift component. To assess the robustness of this identification strategy, we apply the diagnostics proposed by Goldsmith-Pinkham et al. (2020), who show that the shift-share IV estimate can be expressed as a weighted average of just-identified IV estimates, each using a single origin-country share as the instrument. The weight assigned to each share, known as Rotemberg weights and denoted by $\hat{\alpha}_k$ for origin country k, capture the relative contribution of each origin country to the overall identifying variation and are proportional to the first-stage coefficient and the variance of the share instrument across LGAs. This decomposition serves two purposes: it identifies which origin countries drive the aggregate estimate, thereby clarifying the economic content of the instrument, and it allows us to assess the plausibility of the exclusion restriction by examining whether exogeneity is likely to hold for the highest-weighted shares individually.

Table 2 summarizes the Rotemberg weights for the unemployment outcome. Panel A shows that the vast majority of weights are positive, with the sum of positive weights equal to 1.050, accounting for 95.5% of the total weight. This indicates that nearly all origin-country shares contribute positively to the shift-share IV estimate, and that our estimator approximates a convex combination of the just-identified country-specific coefficients $\hat{\beta}_k$, with no indication of misspecification.

[Table 2]

Panel B reports pairwise correlations among the Rotemberg weights ($\hat{\alpha}_k$), the shift component (g_k), the just-identified coefficients ($\hat{\beta}_k$), the first-stage F-statistics ($\hat{F}_k$), and the variance of origin-country shares across LGAs ($\text{Var}(z_k)$). We use the overall unemployment rate as an illustrative outcome; the corresponding tables for other outcomes are reported in Tables A1, A2, and A3. A larger correlation reflects greater relevance of that component of the IV in generating the identifying variation. The results show that the shift component (g_k) has a strong and positive correlation of 0.518 with the Rotemberg weights, whereas the variance of the share component ($\text{Var}(z_k)$) exhibits a high correlation of 0.856, indicating that both components contribute substantially to the identifying variation, with the share component playing a particularly prominent

role. Importantly, the correlation between the Rotemberg weights and the just-identified estimates ($\hat{\beta}_k$) is -0.124, which is modest in magnitude, suggesting that origin countries receiving higher weights do not systematically produce estimates that diverge from the overall IV coefficient. The positive correlation between the weights and the first-stage F-statistics ($\hat{F}_k = 0.262$) further indicates that origin countries contributing more to the identifying variation also tend to serve as stronger individual instruments. Therefore, it is important to verify that those origin-country shares receiving the highest weights are associated with just-identified estimates $\hat{\beta}_k$ broadly consistent with our main estimate, which we turn to in Panel C.

Panel C of Table 2 lists the five origin-country shares receiving the highest Rotemberg weights and, hence, driving most of the identifying variation. The share of immigrants from India explains approximately 33% of the total instrument variation, and the share of immigrants from China generates an additional 8.5%. Together, the top five origin countries—India, China, Philippines, Sri Lanka, and Vietnam—account for a substantial share of the total identifying variation, a pattern broadly consistent with Australia's historical immigration flows. A reassuring feature of our IV is that the just-identified estimates of the highest-weighted shares are, with one exception, broadly consistent in sign with the main estimate. Specifically, India ($\hat{\beta}_k = 0.072$; 95% CI: $0.020, 0.123$) and Sri Lanka ($\hat{\beta}_k = 0.066$; 95% CI: $0.008, 0.124$) each produce positive and statistically significant estimates, with confidence intervals that exclude zero. The Philippines similarly yields a positive estimate ($\hat{\beta}_k = 0.061$), though its confidence interval ($-0.041, 0.164$) marginally includes zero. China's estimate ($\hat{\beta}_k = 0.208$) carries the same sign but is associated with a wide confidence interval ($-0.189, 0.605$), reflecting limited precision in the just-identified specification. Vietnam is the only top-five country whose point estimate is negative ($\hat{\beta}_k = -0.130$), though it is statistically indistinguishable from zero (95% CI: $-0.425, 0.166$) and its Rotemberg weight (0.044) is among the smallest of the top five, limiting its influence on the aggregate estimate.

5 Results

5.1 Main results

Table 3 presents the estimated effects of the immigration share on labor market outcomes using three estimation methods: Pooled OLS, FE, and FE-IV. The dependent variables are the overall unemployment rate, the high-skill employment rate, the middle-skill employment rate, and the low-skill employment rate, reported in Columns (1) through (4), respectively. Panel A in Table 3 reports POLS estimates of the average change in labor market outcomes, in response to changes in the immigration share, controlling for year fixed effects, and LGA fixed effects. The immigration share is positively and signifi-

cantly associated with the overall unemployment rate, suggesting that areas with higher immigrant concentrations exhibit higher unemployment rates on average. Turning to skill-specific employment outcomes, the immigration share bears no statistically significant relationship with high-skill employment in Column 2, while it is positively associated with middle-skill employment and negatively associated with low-skill employment. It is important to note, however, that these estimates do not address the potential endogeneity of the immigration share and should therefore be interpreted as descriptive associations rather than causal effects.

[Table 3]

Panel B of Table 3 reports the FE estimates, which exploit within LGA variation over time to mitigate the influence of time-invariant unobserved characteristics that may confound the relationship between immigration and labor market outcomes. Notably, the sign on the overall unemployment rate reverses relative to the Pooled OLS estimates, becoming negative and statistically insignificant,suggesting that the positive association observed in Panel A is largely driven by cross-sectional sorting of immigrants into high-unemployment areas. Turning to skill-specific employment outcomes, the FE estimates reveal a positive and significant association between the immigration share and high-skill employment, as well as middle-skill employment, while the negative association with low-skill employment persists.

Panel C of Table 3 presents our preferred FE-IV estimates, which address the potential endogeneity of the immigration share while controlling for LGA and year fixed effects. The first-stage results confirm that the shift-share IV is a relevant predictor of the LGA-level immigration share, with a Kleibergen-Paap F-statistic of 8.544, which accounts for heteroskedasticity and clustering in the standard errors. While this falls marginally below the conventional threshold of 10, we complement this diagnostic with Anderson-Rubin F-statistic. The Anderson-Rubin F-statistics range from 8.558 to 20.545 across all four specifications, with all corresponding p-values below 0.01, confirming that the second-stage estimates are robust to weak-instrument concerns. The second-stage estimates reveal heterogeneous effects of the immigration share across labor market outcomes. With the respect to the overall unemployment rate, immigration exerts a positive and statistically significant effect. Turning to skill-specific employment outcomes, the immigration share is positively and significantly associated with high-skill employment, while its effects on middle-skill and low-skill employment are negative and statistically significant.

Compared to the FE estimates in Panel B, the FE-IV estimates exhibit notable differences across specifications. For high-skill employment (Column 2) and low-skill employment (Column 4), the FE-IV estimates are substantially larger in magnitude, suggesting that FE estimation alone underestimates the true impact of immigration on skill-specific

employment outcomes. The overall unemployment rate (Column 1) and middle-skill employment (Column 3) reverse in sign relative to the FE estimates, turning positive and negative, respectively. These patterns collectively suggest that immigrants systematically select into LGAs with relatively favorable labor market conditions—areas with lower unemployment rates and stronger middle-skill employment prospects—thereby introducing a downward bias in the FE estimates. This positive selection underscores the necessity of our IV approach in recovering an unbiased causal estimate.

5.1.1 Labor market outcomes

When we focus on FE-IV estimation for labor market outcomes, our analysis uncovers several notable findings. First, Column (1) documents a positive and significant impact of immigration on the overall unemployment rate. Specifically, a one-standard-deviation increase in the immigration share – approximately 21.5 percentage points – raises the unemployment rate by approximately 2.8 percentage points. Given that the mean unemployment rate across LGAs in our sample is 6.1%, the magnitude of this effect is not small. Turning to skill-specific employment outcomes, Column (2) through (4) present a consistent pattern of heterogeneous effects across the skill distribution. A one-standard-deviation increase in the immigration share raises high-skill employment by approximately 5.7 percentage points, relative to a mean of 35.1%. In contrast, middle-skill and low-skill employment decline by approximately 3.1 and 6.3 percentage points, relative to means of 26.6% and 30.2%, respectively. The magnitude of the negative effect is particularly pronounced for low-skill workers.

This pattern is broadly consistent with the task-based framework of Peri and Sparber (2009), who document that immigrant inflows generate occupational complementarities that reallocate labor toward communication- and language-intensive occupations, producing employment gains at the upper end of the skill distribution. At the same time, the declines in middle- and low-skill employment suggest that this reallocation is not frictionless. Workers displaced from lower-skill occupations face differential barriers to upward mobility: those with sufficient human capital may upgrade into high-skill jobs, contributing to the observed increase in high-skill employment, while workers with limited adaptability are more likely to exit employment altogether, particularly at the lower end of the skill distribution where adjustment frictions are most severe (Dustmann et al., 2017). This mechanism is consistent with the pattern of occupational polarization in which immigration simultaneously raises employment at the top while compressing it in the middle and bottom of the skill distribution (Autor et al., 2003).

The concurrent increase in unemployment, however, suggests that this reallocation is not frictionless. Not all workers displaced from low- and middle-skill occupations are immediately absorbed into higher-skill jobs, consistent with the adjustment cost mechanism

documented by Dustmann et al. (2017), who find that short-run labor market displacement can coexist with longer-run complementarity gains as local labor markets gradually adjust. The magnitude of the low-skill displacement effect—nearly twice that of the middle-skill decline—further suggests that workers at the bottom of the skill distribution face the greatest barriers to occupational mobility, echoing the findings of Borjas (2003) on the concentration of employment losses in low-skill occupations in high-immigration labor markets.

To provide a more nuanced understanding of the aggregate labor market effects documented in Table 3, we examine whether the impacts of immigration differ by gender. Table 4 reports FE-IV estimates separately for male and female workers across the same four labor market outcomes. This disaggregation allows us to assess the extent to which the baseline results mask heterogeneous effects across gender groups, and to identify which segment of the workforce bears a disproportionate share of the labor market adjustment.

[Table 4]

Table 4 reports several noteworthy findings. Panel A indicates that, for male workers, immigration exerts a positive and significant effect on unemployment (0.168) and high-skill employment (0.208), while the effect on low-skill employment is negative and significant (-0.285). The effect on middle-skill employment is negative but statistically insignificant (-0.060). Turning to Panel B, the results for female workers reveal a broadly similar pattern, with immigration raising unemployment (0.099). Comparing the two panels, several gender-specific patterns emerge. The positive effect on unemployment is more pronounced for males (0.168) than for females (0.099), suggesting that males bear a large share of the unemployment burden. In contrast, the negative effect on low-skill employment is larger in magnitude for females than for males, as is the positive effect on high-skill employment, which is larger for females than for males.

Gender-disaggregated estimates reveal that the occupational upgrading pattern documented above holds for both males and females, yet differs meaningfully in its magnitude and distributional consequences. For males, the displacement effect is concentrated almost entirely at the bottom of the skill distribution, with low-skill employment declining significantly while middle-skill employment shows no statistically meaningful response. This suggests that male workers in middle-skill occupations are relatively insulated from immigrant competition, whereas those in low-skill manual and routine jobs bear a disproportionate share of the adjustment burden, consistent with (Autor et al., 2003).

For females, by contrast, significant declines are observed across both middle- and low-skill employment, with the low-skill displacement effect (-0.366) substantially larger than its male counterpart (-0.285). This broader occupational exposure among females may reflect the higher concentration of women in service-oriented middle-skill occupations—such as clerical and sales roles—that are more susceptible to labor market competition

from immigrant inflows, as documented Cortes (2008) in the context of low-skilled immigration and female labor supply. The larger high-skill employment gain among females (0.320) relative to males (0.208) further suggests that the occupational reallocation toward cognitively intensive jobs is more pronounced for women, consistent with a pattern in which female workers face stronger upward reallocation pressure in response to immigrant inflows at the lower end of the skill distribution.

5.1.2 Marriage outcomes

Beyond labor market outcomes, we further examine whether immigration affects marriage decision among young adults. Specifically, the marriage share is constructed as the proportion of married individuals aged 20-34 within each LGA, as this age group represents the primary marriage market cohort and is most likely to be at the margin of marriage formation.Table 5 presents the FE-IV estimates of the immigration share on marriage shares for the overall population, males, and females, reported in Columns (1) through (3), respectively.

[Table 5]

The second-stage estimates reveal that immigration exerts a positive and statistically significant effect on marriage rates across all three specifications. For the overall population, a one-standard-deviation increase in the immigration share raises the marriage share by approximately 8.5 percentage points. Given that the mean marriage share among individuals aged 20-34 is 31.5%, the magnitude of this effect is economically substantial. Disaggregating by gender, the positive effect is statistically significant for both male (0.269) and female (0.520), with the magnitude being considerably larger for females. Specifically, a one-standard-deviation increase in the immigration share raises the male marriage share by approximately 5.8 percentage points relative to a mean of 26.7%, and the female marriage share by approximately 11.2 percentage points relative to a mean of 36.4%. The effect on female marriage share is thus nearly twice as large as that on male marriage share, suggesting that the marriage market consequences of immigration are disproportionately concentrated among young female adults.

Immigration exerts a positive and statistically significant effect on the local marriage share, a finding that warrants careful interpretation in light of the labor market displacement documented above. While the concurrent increase in unemployment might suggest deteriorating economic conditions that would ordinarily suppress marriage formation, the positive effect on marriage share need not be inconsistent with this pattern. As documented by Carlana and Tabellini (2025), the relationship between local labor market conditions and marriage market outcomes need not follow a simple monotonic pattern. In the present context, immigration is associated with rising housing costs in local areas

(Moallemi and Melser, 2020), which may generate additional incentives for household formation: individuals facing housing affordability constraints may find it economically advantageous to pool resources through marriage, particularly when income instability makes independent living less viable (Farnham et al., 2011). This mechanism suggests that immigration reshapes local marriage markets not only through labor market channels but also through its effects on local housing costs.

Gender-disaggregated results reveal that the positive effect is considerably larger for females (0.520) than for males (0.269), a pattern that mirrors the asymmetry documented in the labor market analysis. Recall that female workers experienced broader occupational displacement—spanning both middle- and low-skill jobs—relative to males, for whom displacement was concentrated at the bottom of the skill distribution. This wider dispersion of labor market outcomes among females implies stronger heterogeneity in economic circumstances within the female population, potentially amplifying the gains from marital matching and generating a larger marriage market response, consistent with the notion that greater heterogeneity in economic outcomes strengthens the incentive to form marital unions by raising the returns to positive assortative matching (Becker, 1973)

5.2 Heterogeneous analysis

To more comprehensively understand the effects of immigration on marriage outcomes, we conduct a set of heterogeneity analysis by splitting the sample into different groups. Specifically, we extend our analysis to detect the differential impacts between LGAs with a high proportion of immigrants originating from English-speaking countries and those with a low proportion, as the country of origin may shape the degree of cultural and linguistic proximity to the native population, potentially influencing marriage market outcomes.

Table 6 presents the heterogeneous effects of immigration on marriage outcomes by the English-speaking background of immigrants' country of origin. The results suggest a notable asymmetry between the two groups. In LGAs with a high proportion of immigrants originating from English-speaking countries, the estimated effects on marriage shares are positive but statistically insignificant across all specifications. In contrast, in LGAs with a low proportion of English-speaking origin immigrants, the positive effects on marriage rates are statistically significant across all specifications, for both the overall population and by gender.

[Table 6]

These findings suggest that the positive marriage market effects of immigration are primarily concentrated in LGAs with a higher share of immigrants from non-English-speaking countries, pointing to the role of cultural distance as a potential mediating

factor in shaping marriage market outcomes. The concentration of significant effects in low English-speaking areas is nonetheless economically interpretable and broadly consistent with the mechanisms outlined above. In communities where English proficiency is limited, newly arrived immigrants are more likely to settle within co-ethnic enclaves, reinforcing existing social networks and facilitating marriage formation within tightly knit communities. This is consistent with the enclave hypothesis, whereby linguistic and cultural proximity lowers search frictions in the marriage market and strengthens intra-community matching (Munshi, 2003). The gender asymmetry documented in the aggregate results – females exhibiting a larger marriage response than males – is also preserved within low English-speaking LGAs (0.356 vs. 0.198), suggesting that the income pooling and occupational displacement channels identified earlier operate with particular force in linguistically concentrated communities where female labor market exposure tends to be higher.

6 Robustness checks

We examine the robustness of our main results on labor market outcomes in several ways. As discussed in Section 4.2. the validity of the shift-share IV rests on the exclusion restriction that unobserved factors influencing local labor market conditions are independent of the historical settlement patterns of immigrants across LGAs in 1991. However, recent research has raised concerns about identification challenges with the shift-share IV, particularly when the initial shares used as weights may themselves be endogenous (Jaeger et al., 2018; Goldsmith-Pinkham et al., 2020). In our setting, the variation in the IV across LGAs may partly reflect the immigrant population within each LGA, which contributes a component of the national immigrant population used in constructing the instrument. To assess the sensitivity of our findings to this potential source of endogeneity, we implement a leave-one-out version of the shift-share IV following Goldsmith-Pinkham et al. (2020) and Tabellini (2020), which excludes the immigrants residing in a given LGA when constructing the national-level immigrant count for each country of origin. Table 7 reports the results. The estimated coefficients are broadly consistent with those in Panel C of Table 3 in terms of both sign and statistical significance across all four labor market outcomes, indicating that the potential endogeneity of the initial shares does not pose a serious threat to our identification strategy.

[Table 7]

To further examine whether our results are driven by immigration from specific influential countries of origin, we follow Goldsmith-Pinkham et al. (2020) and compute Rotemberg weights by country of origin to quantify the relative contribution of each origin country to the overall variation in the shift-share IV. These weights are presented

in Table 2. To further examine whether our results are driven by specific influential countries of origin, we follow Goldsmith-Pinkham et al. (2020) and compute Rotemberg weights by country of origin to quantify the relative contribution of each origin country to the overall variation in the shift-share IV, as reported in Table 2. India exhibits the highest Rotemberg weight (0.330), accounting for approximately 33% of the total identifying variation. To assess whether our main findings are sensitive to this concentration, we construct an alternative measure of LGA-level immigration share excluding India from the shift-share instrument (Kim and Wang, 2024). The results remain qualitatively unchanged, confirming that our findings are not driven by immigrants from India alone.

We also explore the robustness of our findings to the inclusion of additional variables. Our baseline specification includes only LGA and year fixed effects along with the lagged share of college graduates as control variable. To account for potential confounding from workforce composition, Table 8 additionally controls for the lagged share of working-age population aged 15 to 64. The estimated coefficients remain consistent with those reported in Panel C of Table 3 across all specifications, confirming that our main results are not sensitive to the inclusion of additional controls.

[Table 8]

7 Concluding remarks

This paper investigates the causal effects of immigration on labor market outcomes and marriage rates in Australia, using census data from the Australian Bureau of Statistics across local government areas (LGAs). To address the endogeneity of the immigration share, we employ a fixed effects instrumental variable (FE-IV) approach with a shift-share IV based on historical immigrant settlement patterns prior to Australia's 1996 immigration policy reform. To assess the credibility of our identification strategy, we follow Goldsmith-Pinkham et al. (2020) and report Rotemberg weights by country of origin, verifying that our findings are not driven by a narrow set of potentially endogenous settlement shares.

Our main results reveal that immigration generates heterogeneous effects on local labor markets. An increase in the immigration share raises the overall unemployment rate and reshapes the skill composition of employment: high-skill employment increases while middle- and low-skill employment decline, consistent with a pattern of occupational upgrading at the upper end of the skill distribution alongside displacement at the lower end. These findings are robust to a battery of robustness checks, including a leave-one-out IV specification and the inclusion of additional control variables, confirming that our results are not driven by the endogeneity of historical settlement patterns or omitted local characteristics.

Disaggregating the main results by gender further reveals that the labor market consequences of immigration differ systematically between male and female workers. The positive effect on high-skill employment and the negative effect on low-skill employment are both larger in magnitude for female workers, while the adverse unemployment effect is more pronounced for males. The negative effect on middle-skill employment is statistically insignificant for males but significant for females, suggesting that female middle-skill workers bear a disproportionate share of the adjustment burden. Turning to marriage outcomes, immigration raises marriage rates among young adults aged 2034, with the effect being substantially larger for females than for males. Our heterogeneity analysis by the English-speaking background of immigrants' country of origin further reveals that the positive marriage market effects are primarily concentrated in LGAs with a higher share of immigrants from non-English-speaking countries.

These findings carry important policy implications. The skill-specific and gender-differentiated nature of immigration's labor market effects suggests that uniform national immigration policies may be insufficient to address the distributional consequences of immigration across local labor markets. In particular, the disproportionate burden borne by low-skill and female workers calls for targeted labor market policies—such as skill upgrading programs and retraining initiatives—that are sensitive to the local skill composition and demographic structure of the workforce. At the same time, the concentration of adverse effects among low-skill workers underscores the importance of place-based support mechasnisms that can cushion the adjustment coasts borne by the most exposed communities. With respect to marriage market outcomes, our findings suggest that immigration reshapes not only the labor market but also the demographic structure of local communities in ways that have broader implications for household formation, family policy, and long-run demographic trends in Australia.

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Tables

Table 1: Summary statistics of immigration share and outcomes

Variables	Mean	St. Dev	Min	Max	N
Panel A. Immigration Exposure					
Immigration share	0.264	0.215	0	2.557	2,976
Panel B. Labor Market Outcomes (Overall)					
Unemployment rate	0.061	0.031	0	0.377	2,976
Employment rate, high skill	0.351	0.104	0.126	0.699	2,976
Employment rate, middle skill	0.266	0.100	0.028	0.493	2,976
Employment rate, low skill	0.302	0.085	0.081	0.722	2,976
Panel C. Labor Market Outcomes (By Gender)					
Unemployment rate (male)	0.064	0.035	0	0.357	2,976
Unemployment rate (female)	0.057	0.028	0	0.385	2,976
Employment rate, high skill (male)	0.353	0.120	0.075	0.811	2,976
Employment rate, high skill (female)	0.350	0.091	0	0.724	2,976
Employment rate, middle skill (male)	0.245	0.083	0	0.485	2,976
Employment rate, middle skill (female)	0.288	0.141	0	0.535	2,976
Employment rate, low skill (male)	0.316	0.077	0.085	0.739	2,976
Employment rate, low skill (female)	0.287	0.121	0.026	0.773	2,976
Panel D. Marriage Outcomes					
Marriage share	0.315	0.094	0	0.703	2,975
Marriage share (male)	0.267	0.082	0	0.709	2,975
Marriage share (female)	0.364	0.116	0	0.788	2,975

Table 2: Summary of Rotemberg Weights - Unemployment Rate

Panel A: Negative and positive weights

	Sum	Mean	Share
$\hat{\alpha}_k \leq 0$	-0.050	-0.007	0.045
$\hat{\alpha}_k > 0$	1.050	0.048	0.955

Panel B: Correlations

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	$\hat{F}_k$	$\text{Var}(z_k)$
$\hat{\alpha}_k$	1.000				
g_k	0.518	1.000			
$\hat{\beta}_k$	-0.124	-0.010	1.000		
$\hat{F}_k$	0.262	0.355	-0.294	1.000	
$\text{Var}(z_k)$	0.856	0.684	-0.034	0.057	1.000

Panel C: Top 5 origin countries

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	95% C.I.	Share
India	0.330	7.7e+04	0.072	(0.020, 0.123)	0.017
China	0.085	1.1e+05	0.208	(-0.189, 0.605)	0.024
Philippines	0.059	9.3e+04	0.061	(-0.041, 0.164)	0.020
Sri Lanka	0.050	4.7e+04	0.066	(0.008, 0.124)	0.010
Vietnam	0.044	1.5e+05	-0.130	(-0.425, 0.166)	0.032

Notes: This table reports statistics about the Rotemberg weights, following Goldsmith-Pinkham et al. (2020). Panel A reports the sum, mean, and the share of negative and positive Rotemberg weights $\hat{\alpha}_k$. Panel B reports correlations between the weights ($\hat{\alpha}_k$), the national component of immigrants (g_k), the just-identified IV estimates ($\hat{\beta}_k$), the first-stage F-statistics ($\hat{F}_k$), and the variance of origin country shares across LGAs ($\text{Var}(z_k)$). Panel C reports the top five origin countries by Rotemberg weight; 95% CIs are $\hat{\beta}_k \pm 1.96 \cdot \hat{se}_k$; Share refers to each origin country's share of total immigrants (g_k) across all origin countries. The residual country group is excluded from this table but included in the estimation.

Table 3: The Effects of Immigration on Labor Market Outcomes

	Unemployment Rate	Employment Rate		
	(1)	(2)	(3)	(4)
	Overall	High-Skill	Middle-Skill	Low-Skill
Panel A. Pooled OLS Estimation				
Immigration share	0.012***	-0.019	0.066***	-0.057***
	(0.004)	(0.023)	(0.014)	(0.013)
LGA fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Control variables	No	No	No	No
Observations	2,976	2,976	2,976	2,976
Panel B. FE Estimation				
Immigration share	-0.003	0.028***	0.032***	-0.054***
	(0.005)	(0.009)	(0.010)	(0.011)
LGA fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Control variables	No	No	No	No
Observations	2,976	2,976	2,976	2,976
Panel C. FE-IV Estimation				
** Second stage*				
Immigration share	0.132***	0.263**	-0.146*	-0.291***
	(0.050)	(0.118)	(0.077)	(0.113)
** First stage*				
IV	0.436***	0.436***	0.436***	0.436***
	(0.149)	(0.149)	(0.149)	(0.149)
C-D F-stat	70.791	70.791	70.791	70.791
K-P F-stat	8.544	8.544	8.544	8.544
A-R F-stat	9.768	20.545	8.558	19.310
A-R p-value	0.002	0.000	0.004	0.000
LGA fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Control variables	Yes	Yes	Yes	Yes
Observations	2,976	2,976	2,976	2,976

Notes: FE: Fixed effects. C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. Robust standard errors clustered at the LGA level are in parentheses. All specifications include LGA and year fixed effects. Control variables are the lagged college graduate share. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 4: The Effects of Immigration on Labor Market Outcomes by Gender

	Unemployment Rate	Employment Rate		
	(1)	(2)	(3)	(4)
	Overall	High-Skill	Middle-Skill	Low-Skill
Panel A. Male				
** Second stage*				
Immigration share	0.168***	0.208**	-0.060	-0.285***
	(0.065)	(0.104)	(0.059)	(0.108)
** First stage*				
IV	0.436***	0.436***	0.436***	0.436***
	(0.149)	(0.149)	(0.149)	(0.149)
C-D F-stat	70.791	70.791	70.791	70.791
K-P F-stat	8.544	8.544	8.544	8.544
A-R F-stat	12.692	12.348	1.368	21.635
A-R *p*-value	0.000	0.000	0.243	0.000
LGA fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Control variables	Yes	Yes	Yes	Yes
Observations	2,976	2,976	2,976	2,976
Panel B. Female				
** Second stage*				
Immigration share	0.099**	0.320**	-0.124*	-0.366***
	(0.048)	(0.138)	(0.074)	(0.136)
** First stage*				
IV	0.436***	0.436***	0.436***	0.436***
	(0.149)	(0.149)	(0.149)	(0.149)
C-D F-stat	70.791	70.791	70.791	70.791
K-P F-stat	8.544	8.544	8.544	8.544
A-R F-stat	4.278	21.836	5.217	21.268
A-R *p*-value	0.039	0.000	0.023	0.000
LGA fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Control variables	Yes	Yes	Yes	Yes
Observations	2,976	2,976	2,976	2,976

Notes: C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. Robust standard errors clustered at the LGA level are in parentheses. All specifications include LGA and year fixed effects. Control variables are the lagged college graduate share. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

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Table 5: The Effects of Immigration on Marriage Share

	(1) Overall	(2) Male	(3) Female
** Second stage*			
Immigration share	0.395***	0.269**	0.520***
	(0.142)	(0.111)	(0.181)
** First stage*			
IV	0.438***	0.438***	0.438***
	(0.149)	(0.149)	(0.149)
C-D F-stat	71.522	71.522	71.522
K-P F-stat	8.628	8.628	8.628
A-R F-stat	24.528	12.420	29.003
A-R p-value	0.000	0.000	0.000
LGA fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Control variables	Yes	Yes	Yes
Observations	2,975	2,975	2,975

Notes: C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. Robust standard errors clustered at the LGA level are in parentheses. All specifications include LGA and year fixed effects. Control variables are the lagged college graduate share. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6: Heterogeneity Analysis by English-Speaking Immigrant Concentration

FE-IV Estimation	High English-Speaking Share			Low English-Speaking Share		
	(1)	(2)	(3)	(4)	(5)	(6)
	Marriage	Male	Female	Marriage	Male	Female
	Share	Marriage	Marriage	Share	Marriage	Marriage
Immigration share	0.443	0.246	0.639	0.276**	0.198**	0.356**
	(0.310)	(0.205)	(0.423)	(0.119)	(0.100)	(0.145)
C-D F-stat	55.726	55.726	55.726	37.021	37.021	37.021
K-P F-stat	2.759	2.759	2.759	6.126	6.126	6.126
A-R F-stat	17.722	5.447	20.974	10.554	5.997	14.344
A-R p-value	0.000	0.020	0.000	0.001	0.015	0.000
LGA fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Control variables	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,457	1,457	1,457	1,518	1,518	1,518

Notes: FE: Fixed effects. C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. Robust standard errors clustered at the LGA level are in parentheses. All specifications include LGA and year fixed effects. Control variables are the lagged college graduate share. Regions are classified as high (low) English-speaking if the share of immigrants from English-speaking countries in 1991 is above (below) the sample median. *** p < 0.01, ** p < 0.05, * p < 0.1.

Table 7: Robustness Checks: Leave-Out IV

	Unemployment Rate	Employment Rate		
	(1)	(2)	(3)	(4)
	Overall	High-Skill	Middle-Skill	Low-Skill
** Second stage*				
Immigration share	0.161**	0.317*	-0.195*	-0.345**
	(0.076)	(0.183)	(0.117)	(0.176)
** First stage*				
Leave-out IV (India)	0.384**	0.384**	0.384**	0.384**
	(0.173)	(0.173)	(0.173)	(0.173)
C-D F-stat	41.322	41.322	41.322	41.322
K-P F-stat	4.914	4.914	4.914	4.914
A-R F-stat	7.274	16.944	8.313	14.956
A-R p-value	0.007	0.000	0.004	0.000
LGA fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Control variables	Yes	Yes	Yes	Yes
Observations	2,976	2,976	2,976	2,976

Notes: C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. The leave-out IV is constructed by excluding India, which has the highest Rotemberg weight, from the shift-share instrument. Robust standard errors clustered at the LGA level are in parentheses. All specifications include LGA and year fixed effects. Control variables are the lagged college graduate share. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8: Robustness Checks: Additional Control

	Unemployment Rate	Employment Rate		
	(1)	(2)	(3)	(4)
	Overall	High-Skill	Middle-Skill	Low-Skill
** Second stage*				
Immigration share	0.139***	0.279**	-0.173**	-0.295**
	(0.053)	(0.124)	(0.080)	(0.118)
Lagged college graduate share	-0.154***	0.254**	0.066	-0.122
	(0.058)	(0.108)	(0.080)	(0.112)
Lagged population share (aged 15–64)	-0.084	-0.193*	0.329***	0.039
	(0.053)	(0.104)	(0.087)	(0.111)
** First stage*				
IV	0.410***	0.410***	0.410***	0.410***
	(0.141)	(0.141)	(0.141)	(0.141)
C-D F-stat	63.612	63.612	63.612	63.612
K-P F-stat	8.444	8.444	8.444	8.444
A-R F-stat	9.577	20.336	12.159	16.357
A-R p-value	0.002	0.000	0.001	0.000
LGA fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	2,976	2,976	2,976	2,976

Notes: C-D F-stat: Cragg-Donald F-statistic. K-P F-stat: Kleibergen-Paap F-statistic. A-R F-stat: Anderson-Rubin F-statistic. Robust standard errors clustered at the LGA level are in parentheses. All specifications include LGA and year fixed effects and the lagged college graduate share. In addition to the baseline controls, this specification includes the lagged share of the population aged 15 to 64. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix

Table A1: Summary of Rotemberg Weights - High-Skill Employment Rate

Panel A: Negative and positive weights

	Sum	Mean	Share
$\hat{\alpha}_k \leq 0$	-0.050	-0.007	0.045
$\hat{\alpha}_k > 0$	1.050	0.048	0.955

Panel B: Correlations

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	$\hat{F}_k$	$\text{Var}(z_k)$
$\hat{\alpha}_k$	1.000				
g_k	0.518	1.000			
$\hat{\beta}_k$	-0.048	-0.074	1.000		
$\hat{F}_k$	0.262	0.355	-0.301	1.000	
$\text{Var}(z_k)$	0.856	0.684	-0.005	0.057	1.000

Panel C: Top 5 origin countries

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	95% C.I.	Share
India	0.330	7.7e+04	0.152	(0.041, 0.263)	0.017
China	0.085	1.1e+05	0.299	(-0.526, 1.125)	0.024
Philippines	0.059	9.3e+04	0.412	(-0.013, 0.838)	0.020
Sri Lanka	0.050	4.7e+04	0.138	(-0.013, 0.289)	0.010
Vietnam	0.044	1.5e+05	0.531	(-0.441, 1.503)	0.032

Notes: This table reports statistics about the Rotemberg weights, following Goldsmith-Pinkham et al. (2020). Panel A reports the sum, mean, and the share of negative and positive Rotemberg weights $\hat{\alpha}_k$. Panel B reports correlations between the weights ($\hat{\alpha}_k$), the national component of immigrants (g_k), the just-identified IV estimates ($\hat{\beta}_k$), the first-stage F-statistics ($\hat{F}_k$), and the variance of origin country shares across LGAs ($\text{Var}(z_k)$). Panel C reports the top five origin countries by Rotemberg weight; 95% CIs are $\hat{\beta}_k \pm 1.96 \cdot \hat{se}_k$; Share refers to each origin country's share of total immigrants (g_k) across all origin countries. The residual country group is excluded from this table but included in the estimation.

Table A2: Summary of Rotemberg Weights - Middle-Skill Employment Rate

Panel A: Negative and positive weights

	Sum	Mean	Share
$\hat{\alpha}_k \leq 0$	-0.050	-0.007	0.045
$\hat{\alpha}_k > 0$	1.050	0.048	0.955

Panel B: Correlations

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	$\hat{F}_k$	$\text{Var}(z_k)$
$\hat{\alpha}_k$	1.000				
g_k	0.518	1.000			
$\hat{\beta}_k$	0.101	-0.018	1.000		
$\hat{F}_k$	0.262	0.355	0.240	1.000	
$\text{Var}(z_k)$	0.856	0.684	0.006	0.057	1.000

Panel C: Top 5 origin countries

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	95% C.I.	Share
India	0.330	7.7e+04	-0.048	(-0.116, 0.020)	0.017
China	0.085	1.1e+05	-0.419	(-1.310, 0.471)	0.024
Philippines	0.059	9.3e+04	-0.154	(-0.379, 0.071)	0.020
Sri Lanka	0.050	4.7e+04	-0.034	(-0.091, 0.023)	0.010
Vietnam	0.044	1.5e+05	-0.096	(-0.310, 0.118)	0.032

Notes: This table reports statistics about the Rotemberg weights, following Goldsmith-Pinkham et al. (2020). Panel A reports the sum, mean, and the share of negative and positive Rotemberg weights $\hat{\alpha}_k$. Panel B reports correlations between the weights ($\hat{\alpha}_k$), the national component of immigrants (g_k), the just-identified IV estimates ($\hat{\beta}_k$), the first-stage F-statistics ($\hat{F}_k$), and the variance of origin country shares across LGAs ($\text{Var}(z_k)$). Panel C reports the top five origin countries by Rotemberg weight; 95% CIs are $\hat{\beta}_k \pm 1.96 \cdot \hat{se}_k$; Share refers to each origin country's share of total immigrants (g_k) across all origin countries. The residual country group is excluded from this table but included in the estimation.

Table A3: Summary of Rotemberg Weights - Low-Skill Employment Rate

Panel A: Negative and positive weights

	Sum	Mean	Share
$\hat{\alpha}_k \leq 0$	-0.050	-0.007	0.045
$\hat{\alpha}_k > 0$	1.050	0.048	0.955

Panel B: Correlations

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	$\hat{F}_k$	Var(z_k)
$\hat{\alpha}_k$	1.000				
g_k	0.518	1.000			
$\hat{\beta}_k$	0.022	0.045	1.000		
$\hat{F}_k$	0.262	0.355	0.156	1.000	
Var(z_k)	0.856	0.684	0.027	0.057	1.000

Panel C: Top 5 origin countries

	$\hat{\alpha}_k$	g_k	$\hat{\beta}_k$	95% C.I.	Share
India	0.330	7.7e+04	-0.183	(-0.294, -0.071)	0.017
China	0.085	1.1e+05	-0.150	(-0.612, 0.313)	0.024
Philippines	0.059	9.3e+04	-0.371	(-0.742, 0.000)	0.020
Sri Lanka	0.050	4.7e+04	-0.175	(-0.325, -0.025)	0.010
Vietnam	0.044	1.5e+05	-0.299	(-0.789, 0.191)	0.032

Notes: This table reports statistics about the Rotemberg weights, following Goldsmith-Pinkham et al. (2020). Panel A reports the sum, mean, and the share of negative and positive Rotemberg weights $\hat{\alpha}_k$. Panel B reports correlations between the weights ($\hat{\alpha}_k$), the national component of immigrants (g_k), the just-identified IV estimates ($\hat{\beta}_k$), the first-stage F-statistics ($\hat{F}_k$), and the variance of origin country shares across LGAs (Var(z_k)). Panel C reports the top five origin countries by Rotemberg weight; 95% CIs are $\hat{\beta}_k \pm 1.96 \cdot \hat{se}_k$; Share refers to each origin country's share of total immigrants (g_k) across all origin countries. The residual country group is excluded from this table but included in the estimation.